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RNAlater Recipe

Forked from RNAlater Recipe

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Abstract

This is a storage solution that stabilizes and protectsRNA while inactivating RNase

Guidelines

Note: If you read the patent application, they refer to an ammonium sulfate concentration of 700g per L. What they really mean is add 700g to 1L (including EDTA and sodium citrate at conc. listed above), so you end up with more than 1L total volume.

Note: The pH should be ~7 using this recipe. The patent states that pH can range from 4-8; the preferred commercial pH is 5.2; I use pH 7 for bacterioplankton samples, but pH can be lowered by adding H_2SO_4.

Note: Since this is a storage solution for RNA, use high quality reagents ("molecular biology" grade) when preparing the buffer.

Protocol materials

Ammonium sulfate **Catalog #**97061-184

Troubleshooting

Preparation

1 Prepare 0.5 M EDTA

Protocol

NAME

0.5M EDTA

CREATED BY

Bonnie Poulos

Preview

1.1 Add 186.1 g EDTA disodium salt dihydrate (Mallinckrodt 4931) to 700 ml MilliQ water

1.2 Stir to dissolve

1.3 Bring pH to 8.0 with NaOH (~50ml of 10M NaOH or ~23g NaOH pellets)

1.4 Bring volume to 1L with MilliQ water

1.5 Autoclave

2 Prepare 1 M Sodium citrate

Protocol

NAME

1M Sodium citrate

CREATED BY

Bonnie Poulos

Preview

2.1 Add 294 g Na Citrate dihydrate (Mallinckrodt 0754-12) to 700ml MilliQ water

2.2 Stir to dissolve

2.3 Bring volume to 1L with MillQ water

2.4 Autoclave

RNAlater recipe

3 In a 2 or 3 L flask mix 1,400 mL MilliQ water with 60 mL 0.5M EDTA and 37.5 mL 1M sodium citrate with stirring

4 While continuing to stir the solution, add 1.05 kg (= 1,050 g) Ammonium sulfate in 100 g amounts.

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5 Stir ~30 min on hot plate set to low

00:30:00

Note

It is possible that a tiny amount ammonium sulfate will not go into solution. This is okay; the most important thing is to have a saturated ammonium sulfate solution. Crystals should dissolve in ~30min.

6 Allow solution to cool at room temperature for at least 30min, but overnight on bench is best

00:30:00

7 Filter solution through 0.2 µm filter to remove particles

8 Store at room temp or at 4°C